Redundancy, Geography, and the Supply-Network Precipice:
A Directed k-Core View of Multisourcing under Correlated Regional Shocks

Arjun Saini Aryavardhan Singh

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Abstract

When does the failure of a few firms or regions cascade into a collapse of production? Elliott, Golub, and Leduc [1] (EGL) predict a discontinuous "precipice" in output; Carvalho, Nirei, Saito, and Tahbaz-Salehi [2] (CNST) find that the 2011 Japanese earthquake propagated smoothly; and Dorogovtsev, Goltsev, and Mendes [3] (DGM) show that requiring several surviving neighbors makes network cores vanish abruptly. None of them asks how redundant sourcing interacts with geographically correlated shocks. We extend EGL so that a firm needs k of its n suppliers per input, with suppliers grouped into regions that fail together, and show that this is a directed k-core. We prove that collapse is abrupt exactly when $km \geq 2$ (m = inputs per firm); that co-located and fully spread suppliers have thresholds linked by the closed form $\phi^*_{\text{co}} = 1-(1-\phi^*_{\text{spread}})^{1/m}$, so with one input geography does not move the threshold at all; and that splitting supplier clusters further never hurts when $k = 1$. For $k \geq 2$, however, we find a *diversification trap*: splitting suppliers into a few regional clusters can be worse than both full co-location and full spreading. Simulations on networks of 420,000 firms confirm every prediction, including the trap, and reproduce DGM's thresholds.

1 Introduction

In a supply network a cascade has a direct economic cost: a firm missing one essential input cannot produce, and its customers then face the same problem. The question joins two course units, Cascading Behavior (threshold rules on neighbors) and Random Graphs (branching processes, phase transitions, the giant component). We ask how *redundancy* (needing several working suppliers) and *geography* (suppliers that fail together) move the point at which a supply network collapses. We anchored our search on EGL, followed papers citing it from 2021 to 2026, and scored candidates on relevance, impact, recency, and whether they support analysis and simulation. CNST is the most cited empirical companion, and its shock is regional. We added DGM because it supplies the mathematics of abrupt collapse that EGL's precipice and our model rely on. None of the three is linked from the course page. Sections 2–4 analyze the papers, Section 5 connects them, Section 6 develops our model and results, and Section 7 tests them on simulated networks.

2 Paper 1: Elliott, Golub, and Leduc (2022), "Supply Network Formation and Fragility"

Technical content. Firms sit in deep random supply trees. A firm at depth $d > 0$ needs m essential inputs and has n potential suppliers at depth $d-1$ for each; each relationship works independently with probability x (*relationship strength*), and a firm is functional if every input has at least one working, functional supplier. If a supplier works with probability r,

$$r_d = R_x(r_{d-1}), \qquad R_x(r) = \left(1-(1-xr)^n\right)^m, \tag{1}$$

and deep networks converge to the largest fixed point. For $m \geq 2$, R_x is convex then concave, so as x falls the largest crossing with the diagonal disappears at a tangency and reliability *jumps* to zero at a critical x_{crit} (their Proposition 1); with $m = 2$, $n = 4$, $x_{\text{crit}} = 0.518$ and reliability drops from 0.64 to 0. For $m = 1$ the transition is continuous at $x = 1/n$. EGL then let each firm pay to raise its own x above a baseline $\underline{x}$. Because each firm's investment substitutes for its suppliers' near equilibrium, for $m \geq 3$ a whole interval of cost parameters puts the equilibrium within ε of x_{crit} (their Theorem 1), where an arbitrarily small drop in $\underline{x}$ destroys production (their Proposition 3), while a planner never chooses x near x_{crit}.

Course connections, weaknesses, directions. The rule is a deterministic threshold rule (an AND across inputs of an OR across suppliers), the mirror image of the adoption cascades of Granovetter [7] and Easley–Kleinberg [6, §19.7]; with $m \geq 2$ it is a complex contagion [4]. With $m = 1$ it is the survival of a Galton–Watson tree [6, §21.8], the mechanism behind the giant component. The novelty, in the "economic issues" view of cascades [9], is that strategic

play sets the threshold. The main weakness for us is that link failures are independent and the only aggregate shock is a uniform drop in x, whereas real shocks are regional, so co-located suppliers fail together. The "at least one supplier" rule also ignores capacity, the tree has no hubs or cycles, the main theorem needs $m \geq 3$, and x_{crit} is unobservable. We address the first two.

3 Paper 2: Carvalho, Nirei, Saito, and Tahbaz-Salehi (2021), "Supply Chain Disruptions"

Technical content. The 2011 earthquake hit four prefectures with 4.6% of Japan's output; national growth fell about 0.43 percentage points (pp), far more than their direct share. On 750,237 firms with binary supplier–customer links (about 5 suppliers each, Pareto degree tails), a difference-in-differences design shows sales growth fell 3.8pp for firms with disaster-area suppliers and 3.1pp for firms with disaster-area customers, decaying with network distance and robust to single-plant samples, the split power grid, and a rewired placebo network. A nested-CES model, with complementary primary and intermediate inputs ($\sigma \approx 0.60$) and mildly substitutable intermediates ($\xi \approx 1.18$), attributes 0.47pp of lost GDP growth to the disaster, versus 0.21pp without links out of the region.

Course connections, weaknesses, directions. This is a clean causal measurement of contagion, and its pattern (decay per hop) resembles an attenuating simple contagion. The Leontief inverse $\sum_{r\geq 0}(\mu A)^r$ is a damped sum over walks, the structure of Katz centrality and PageRank [6, §14.6]. But the model is linearized around Cobb–Douglas, so it cannot produce a threshold; exposure is minimum network distance, which treats 1 of 10 suppliers in the disaster area like 10 of 10; and the panel drops firms that exited. Section 8 turns our results into a test that fixes these gaps.

4 Paper 3: Dorogovtsev, Goltsev, and Mendes (2006), "k-Core Organization of Complex Networks"

Technical content. The k-core is the largest subgraph in which every vertex has at least k neighbors, found by repeatedly pruning vertices of degree below k. DGM study it in the configuration model with degree distribution $P(q)$ after a random fraction $Q = 1 - p$ of vertices is removed. Because the graph is locally tree-like, a vertex is in the core if at least k neighbors root infinite $(k-1)$-ary subtrees. With R the probability that an edge does not lead to such a root, $R = 1 - p + p\,\Phi_k(R)$, where Φ_k is built from the excess-degree generating function, and the core first exists where $1/p$ touches the maximum of $[1-\Phi_k(R)]/(1-R)$. For $k = 2$ this is ordinary percolation, a continuous transition. For $k \geq 3$ (with finite second moment) it is *hybrid*: the core appears with a jump plus a square-root singularity, and the clusters of core vertices with exactly k core neighbors (the "corona") diverge in size. On an Erdős–Rényi graph with mean degree 10, the highest (7-)core vanishes after removing only 1.2% of vertices, while the giant component survives up to 90%. For heavy-tailed networks with exponent $\gamma \leq 3$ the cores become ultra-robust to random damage.

Course connections, weaknesses, directions. The analysis is the configuration model with generating functions from the Random Graphs unit, with the giant component as its $k = 2$ case, and pruning a core is a threshold cascade; dense clusters are exactly what block adoption cascades [6, Ch. 19]. The model is undirected and uncorrelated, damage is random and independent, every vertex has the same threshold, and nothing explains why a network would sit near its threshold. Supply networks need the opposite of each: directed dependence on suppliers, correlated regional damage, and economic incentives. Our model supplies the first two, and EGL supply the third.

5 Synthesis

DGM give the mathematics (needing two or more surviving neighbors makes collapse abrupt), EGL the economics (free-riding pushes networks toward that edge), and CNST the evidence and the shock that matters (large and regional), yet CNST's propagation is smooth, which is consistent with a Japanese network that was *not* critical in 2011. None covers how redundancy interacts with correlated damage: EGL protect firms by multisourcing but shocks are independent, DGM's damage is random, and CNST's model has no redundant suppliers. Our model fills this gap and, as we show next, is a directed version of DGM's k-core.

6 Model and Results: k-of-n Multisourcing with Regional Shocks

Setup. We keep EGL's trees and make two changes. First, an input is sourced only if at least k of its n suppliers are both functional and linked by a working relationship, which models capacity limits or dual-sourcing mandates ($k = 1$ is EGL). Second, there are many regions, each hit independently with probability ϕ, and every firm in a hit region

fails. A firm's n suppliers for each input form c clusters of $b = n/c$; a cluster shares one region, and clusters sit in independent regions ($c = n$: fully spread; $c = 1$: co-located). Let $S_k(N,p) = \Pr[\mathrm{Bin}(N,p) \geq k]$, $s = 1-\phi$, and r the probability that a deep firm works given that its own region survives. An input keeps $C \sim \mathrm{Bin}(c,s)$ clusters, hence Cb suppliers, each usable with probability xr, so

$$r = F(r) := \Big[\sum_{C=0}^{c} \tbinom{c}{C} s^C \phi^{c-C} S_k(Cb, xr)\Big]^m, \qquad \rho = s\,r^*, \tag{2}$$

where r^* is the largest fixed point and ρ the aggregate reliability. x_{crit} is the smallest x with $r^* > 0$ when $\phi = 0$, and ϕ^* the largest ϕ with $r^* > 0$. With $m = 1$, $c = n$, and $u = sr$, (2) reads $u = sS_k(n, xu)$, the same form as DGM's Erdős–Rényi equation $u = p\Pr[\mathrm{Poisson}(zu) \geq K-1]$ with core index $K = k+1$: our model is a *directed k-core*, where firms survive only if k *suppliers* survive.

Theorem 1 (Abrupt iff $km \geq 2$)**.** *Let $n \geq \max(k,2)$. The collapse of r^* is discontinuous if and only if $km \geq 2$.*

Proof. If $km \geq 2$, a union bound gives $S_k(N,p) \leq \binom{n}{k} p^k$, so $F(r) \leq Cr^{km} \leq Cr^2$ for a constant C independent of x and ϕ. Thus $F(r) < r$ on $(0,\varepsilon)$ with $\varepsilon = 1/C$, so every positive fixed point is at least ε; since $r^* = 1$ at $x = 1, \phi = 0$ (as $n \geq k$) and $r^* = 0$ for small x, r^*, monotone in the parameters, must jump. If $k = m = 1$, F is strictly concave (as $n \geq 2$) with $F'(0) = nsx$: a branching process whose survival grows continuously once $nsx > 1$. □

This unifies EGL's precipice ($m \geq 2$) and DGM's hybrid transition ($K = k+1 \geq 3$): what matters is needing two or more things to work at once, whether different inputs or several suppliers of one input. With $m = 1$, $n = 4$, $k = 2$, $x_{\mathrm{crit}} = 0.725$ and reliability jumps to 0.745.

Proposition 2 (Spreading turns regional shocks into systemic ones)**.** *If $c = n$, then $F(r) = [S_k(n, sxr)]^m$ by binomial thinning, so a regional shock is exactly a drop in strength from x to sx, and $\phi^*_{\mathrm{spread}} = 1 - x_{\mathrm{crit}}/x$ for $x \geq x_{\mathrm{crit}}$.*

Theorem 3 (Closed form for the two extremes)**.** *For $x \geq x_{\mathrm{crit}}$, $\phi^*_{\mathrm{co}} = 1 - (1-\phi^*_{\mathrm{spread}})^{1/m} = 1 - (x_{\mathrm{crit}}/x)^{1/m}$, where "co" is $c = 1$.*

Proof. Let $G(u) = [S_k(n,xu)]^m$. For $c = n$, $r = G(sr)$, i.e. $u = sG(u)$ with $u = sr$; for $c = 1$, $r = s^m G(r)$. A positive fixed point of $u = aG(u)$ exists iff $a \geq a^* := 1/\sup_u G(u)/u$ (strictly, if $km = 1$; the threshold is the same), so $s^*_{\mathrm{spread}} = a^*$ and $(s^*_{\mathrm{co}})^m = a^*$. □

So with one input ($m = 1$), co-located and spread suppliers collapse at *exactly* the same shock (spreading only raises reliability below it, by the factor $1/s$), while with m inputs co-location cuts the tolerable shock by roughly a factor of m: correlation hurts through complexity. For $m = 2$, $n = 4$, $x = 0.9$ the formula maps $0.425 \to 0.242$ ($k = 1$) and $0.077 \to 0.039$ ($k = 2$), matching the numerical thresholds. Redundancy is costly on its own: requiring $k = 2$ instead of $k = 1$ raises x_{crit} from 0.518 to 0.831, and it takes $n = 8$ potential suppliers to bring it back to 0.500. The collapse also gives little warning: with $k = 2$ and spread suppliers, reliability is still 0.91 with 5% of regions hit and zero past 7.7%.

Theorem 4 ($k = 1$: spreading never hurts)**.** *If $k = 1$ and $c \mid c'$, then $F_c \leq F_{c'}$ pointwise, so r^* and ϕ^* are weakly larger for the finer split c'.*

Proof. An input succeeds with probability $\mathbb{E}[h(B_c)]$, where $B_c = b\,\mathrm{Bin}(c,s)$ counts surviving suppliers and $h(N) = 1-(1-xr)^N$ is concave. Within a cluster, $b\,\mathrm{Bern}(s)$ is the comonotonic sum of b/b' copies of $b'\,\mathrm{Bern}(s)$, so it dominates their independent sum in convex order [5]; convex order is preserved under independent sums [10], so $B_c \geq_{cx} B_{c'}$ and $\mathbb{E}h(B_c) \leq \mathbb{E}h(B_{c'})$. □

Proposition 5 (The diversification trap)**.** *Theorem 4 does not extend to $k \geq 2$: for some parameters, an intermediate split is strictly worse than both extremes.*

With $n = 4$, $k = 2$, $m = 1$, $x = 0.9$: $\phi^* = 0.1946$ when co-located, 0.1705 for two regional pairs, and 0.1946 when fully spread. The convex-order argument breaks because for $k \geq 2$, $S_k(N,p)$ is S-shaped in N (convex, then concave), so convex order no longer ranks the splits. Concretely, losing one of two pairs leaves exactly k suppliers, both of which must work, so firms become "barely sourced" (DGM's corona). The trap also occurs with two inputs ($n = 6$, $k = 3$, $m = 2$: $\phi^* = 0.0529$ co-located versus 0.0461 for two clusters). Scanning 360 parameter combinations ($n \in \{2,4,6,8\}$, all $k \leq n$, $m \leq 3$, six values of x), splitting clusters lowers ϕ^* in 72, all with $k \geq 2$. The policy lesson is model-specific but sharp: a network whose firms each need two working suppliers per input can be *more fragile* when every firm splits its suppliers between two regions than when every firm keeps them together.

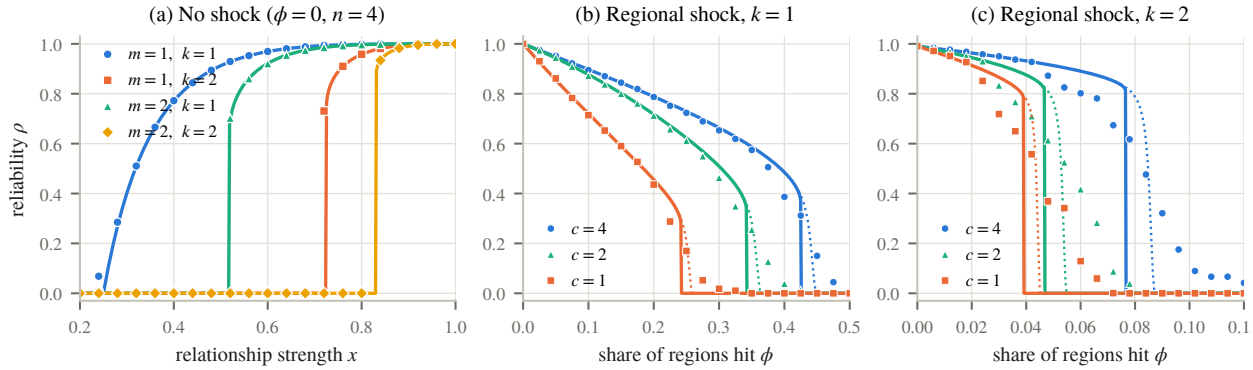


Figure 1: Simulation (markers) against theory (solid: infinite depth; dotted: depth 20). (a) Reliability against relationship strength: $k = 2$ creates a jump even with one input (Theorem 1). (b, c) Reliability against the share of regions hit ($m = 2$, $n = 4$, $x = 0.9$).

Table 1: Collapse threshold ϕ^* ($n = 4$, $x = 0.9$). Simulated thresholds are the realized-shock split point that best separates collapsed runs ($\rho < 0.1$) from survivors, with 95% bootstrap intervals from 270 runs per row (resolution 0.0025, since $G = 200$). The last column is Theorem 3 applied to the deep spread threshold.

m	k	Geography c	Theory (deep)	Theory (depth 20)	Simulation [95% CI]	Closed form
2	1	4 (spread)	0.425	0.444	0.445 [0.440, 0.450]	–
2	1	1 (co-located)	0.242	0.254	0.255 [0.253, 0.258]	0.242
2	2	4 (spread)	0.077	0.086	0.085 [0.085, 0.087]	–
2	2	2	0.047	0.054	0.053 [0.053, 0.055]	–
2	2	1 (co-located)	0.039	0.045	0.042 [0.042, 0.045]	0.039
1	2	4 (spread)	0.195	0.216	0.218 [0.218, 0.218]	–
1	2	2 (trap)	0.171	0.195	0.195 [0.193, 0.198]	–
1	2	1 (co-located)	0.195	0.217	0.218 [0.218, 0.220]	0.195

7 Experiments on Simulated Networks

Setup and validation. The results above are mean-field results for infinitely deep trees, so we test them on finite networks: 21 layers of $N = 20{,}000$ firms over $G = 200$ regions, where each firm draws, for each input, c regions and b distinct suppliers from the previous layer in each; one random set of hit regions applies to all layers. Since only 200 regions exist, we compare each run to the theory at its realized share of hit regions, in the deep limit and at depth 20. Over 3,360 runs (Figure 1), the mean absolute error against depth-20 theory is 0.0025 (0.0010 away from thresholds); visible gaps sit just past each threshold, where 20 layers are too few to reach zero. In total we ran 10,680 seeded simulations.

Abruptness (Theorem 1). At each configuration's theoretical ϕ^* we ran 300 single-input networks (Figure 2a). With $k = 1$, 90% of runs end below 0.1 (maximum 0.21), a continuous fade; with $k = 2$ the outcomes split: 20% collapse below 0.1 and 71% survive above 0.4 (median 0.67). A little extra bad luck in which regions are hit flips the whole network.

Geography (Theorems 3–4, Proposition 5). Table 1 reads ϕ^* directly from the simulations. Every simulated threshold lies within 0.003 of the depth-20 theory, and the closed form reproduces both co-located thresholds. With two inputs, co-location costs tolerance; with one input ($m = 1$) the two extremes collapse together, and the two-pair split collapses *earlier*: its interval lies strictly below both. Figure 2(c) shows all four configurations we tested: the trap appears with one input ($n = 6$, $k = 3$: 0.153 for two clusters versus 0.203 and 0.200 at the extremes) and with two inputs ($n = 6$, $k = 3$, $m = 2$: 0.053 versus 0.058 co-located), while for $m = 3$, $k = 2$ the ordering is monotone. Across all 14 configurations, the depth-20 theory lies within 0.003 of the simulated interval.

Realism. Giving suppliers heavy-tailed popularity (Pareto shape 1.25, as in CNST's degree tails) changed mean reliability by at most 0.002 ($k = 1$) and 0.03 ($k = 2$, within noise). Concentrating *supply* geographically mattered

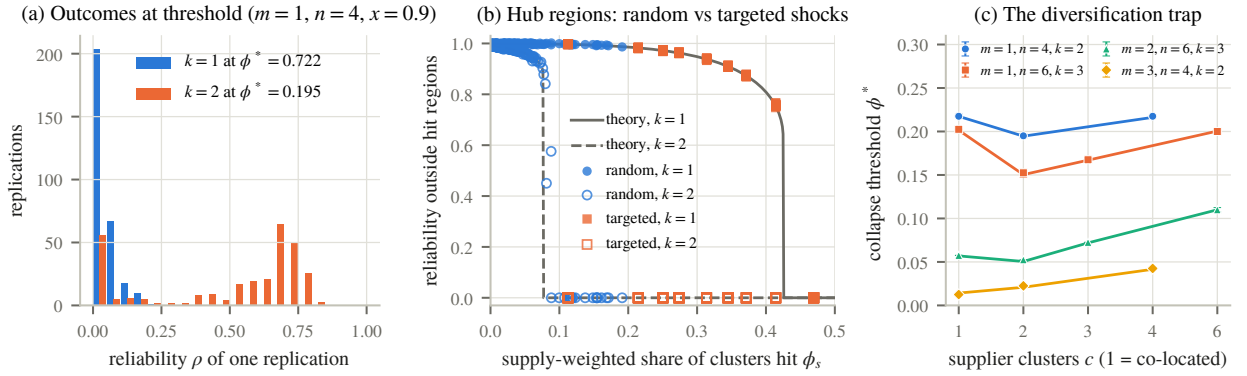


Figure 2: (a) 300 replications at the theoretical threshold ($m = 1$): $k = 1$ fades, $k = 2$ is bimodal. (b) Supply concentrated in hub regions ($m = 2$, $n = 4$, $c = 4$, $x = 0.9$): runs fall on the theory curve once the shock is measured by the supply-weighted share of hit clusters. (c) The diversification trap: depth-20 theory (lines) and simulated ϕ^* with 95% intervals against the number of supplier clusters.

far more: when the largest region hosts 11% of supply relationships but 0.5% of firms, hitting it alone collapses a $k = 2$ network, while hitting up to 10 random regions (5% of firms) collapses it in only 5% of runs, and every run falls on the theory curve once ϕ is replaced by the supply-weighted share of hit clusters (Figure 2b; error 0.002). Finally, our simulator reproduces DGM: on Erdős–Rényi graphs with 100,000 vertices and mean degree 10, each k-core ($k = 3, \ldots, 7$) is present 0.01 below its predicted threshold and gone 0.01 above it, and the 7-core falls from 73.5% of vertices to zero as removal goes from 0.2% to 2.2%.

8 Discussion, Further Research, and Conclusion

Our results sharpen what CNST's data could test. Because exposure should be measured per input and per region, not by network distance, re-estimating CNST's design with the share of each firm's suppliers (by input) in the disaster area should show a kink where fewer than k suppliers survive and bimodal outcomes or exit spikes among the most exposed firms; for industries whose firms need several suppliers per input, the trap predicts that sourcing from a few regional clusters can be more fragile than both co-located and fully spread sourcing. On the theory side, a characterization of when the trap occurs, endogenous geography (EGL's reliability externality suggests firms under-diversify, and the trap suggests they may even choose the worst split), and configuration-model graphs with cycles are the next steps. Choosing firms to harden under a budget is a reverse influence-maximization problem [8] whose AND structure likely breaks submodularity.

In sum, abrupt collapse requires only that firms need two things at once ($km \geq 2$), geography affects it through a closed form that makes co-location costly exactly when production is complex, and the folk wisdom "spread your suppliers" holds for $k = 1$ but can backfire for capacity-constrained buyers, for whom, in our model, a partial split can be the worst choice. Simulations confirm each prediction, and a region holding 0.5% of firms but 11% of supply can collapse a network on its own.

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